

Electron Configuration

Answer key with worked solutions

A. Understanding

1. p subshell.

3 orbitals, each holding 2 electrons: **6 electrons** in total.

2. Pauli and Hund.

Pauli: an orbital holds at most two electrons, and they must have opposite spins. **Hund:** electrons in the same subshell occupy separate orbitals with parallel spins before any orbital is doubly occupied.

B. Atoms

3. Full configurations.

(a) C: $1s^2 2s^2 2p^2$ (b) Mg: $1s^2 2s^2 2p^6 3s^2$ (c) Cl: $1s^2 2s^2 2p^6 3s^2 3p^5$

4. Noble-gas configurations.

(a) Ca: **[Ar] $4s^2$** (b) Fe: **[Ar] $3d^6 4s^2$** (c) Br: **[Ar] $3d^{10} 4s^2 4p^5$**

5. Identify the elements.

(a) 7 electrons: **nitrogen** (b) $10 + 7 = 17$: **chlorine** (c) $18 + 14 = 32$: **germanium**

C. Orbital diagrams

6. Nitrogen.

2s: $\uparrow\downarrow$ 2p: $\uparrow$ $\uparrow$ $\uparrow$. By Hund's rule the three 2p electrons occupy separate orbitals: **3 unpaired electrons**.

7. Unpaired electrons.

(a) S, **[Ne] $3s^2 3p^4$** : 3p is $\uparrow\downarrow$ $\uparrow$ $\uparrow$, so **2** (b) Ne, $1s^2 2s^2 2p^6$: all subshells full, so **0**

D. Ions

8. Ions.

(a) O^{2-} : $8 + 2 = 10$ electrons, $1s^2 2s^2 2p^6 = [\text{Ne}]$ (b) Mg^{2+} : $12 - 2 = 10$ electrons, $[\text{Ne}]$ (c) Fe^{3+} : remove both 4s electrons, then one 3d: **$[\text{Ar}] 3d^5$** . O^{2-} **and** Mg^{2+} have the same configuration (they are isoelectronic with neon).

E. Challenge

9. Chromium.

A half-filled d subshell (one electron in each of the five 3d orbitals) is especially stable, so one 4s electron moves into 3d. Unpaired electrons: five in 3d plus one in 4s = **6**.